

- you want colour temperature/scene control
- you want minimal electrician involvement

Route B: Smart switches / smart relays behind the plate

Best for: families (physical switch still works), owner setups

Downside: depends heavily on wiring (neutral, depth of back box, load types)

Use smart switches/relays when:

- you want wall control + app + automation together
- you can commit to proper installation (preferably by an electrician)

Route C: Smart “modules” only for selected circuits

Best for: phased upgrades, budget-limited builds

Downside: mixed experience across rooms (some smart, some not)

Pro tip: Start with Route A or C for Phase 1, then move to Route B after you’ve confirmed wiring feasibility and your family’s comfort.

Placing smart devices in the home

1) Motion sensors

Ideal: passages, bathrooms, entry corridor, kitchen at night

Avoid: directly facing windows/curtains, ceiling fans at full wobble, areas with pets (unless sensor supports pet immunity)

Step-by-step: Motion sensor placement test

1. Stick it temporarily (tape/3M strip) at chest height.
2. Walk normal paths for 24 hours.
3. Note false triggers and blind spots.
4. Then commit to mounting.

2) Door/window sensors

Ideal: main door, balcony door, windows that are frequently opened



Tip: install on the “still” frame side + moving shutter side with tight alignment.

3) Hubs/controllers

Put your “brain” in a place that’s:

- central (not in a corner bedroom)
- elevated (not on the floor behind a TV unit)
- on stable power (UPS/inverter if possible)

This matters even more in concrete-heavy homes.

Earthing and leakage protection

Two safety concepts show up repeatedly in Indian standards and practice:

1. **Earthing:** safe fault current paths and reducing shock risk
2. **Automatic disconnection/residual current protection:** disconnecting supply when leakage is detected

Indian wiring practice references such coordination of protective devices and earthing arrangements for fault conditions in standards like IS 732 (code of practice for electrical wiring installations). Earthing as a consolidated design/installation/maintenance topic is covered in IS 3043 (code of practice for earthing).

Practical takeaway

If your home is older, renovated multiple times, or has frequent nuisance trips/shocks:

- get earthing checked by a qualified electrician
- consider updating protection at the DB (RCCB/RCBO selection and correct wiring should be done professionally)

Some Indian safety references also explicitly call out household usage of RCCBs with 30 mA sensitivity (as cited via National Electrical Code references in official training material).

Build your “room automation plan”

For each room, fill this:

1. **Primary action** (what should happen automatically?)
Example: “Passage light turns on after sunset when motion detected.”
2. **Manual override** (how will a human control it?)
Example: wall switch still works / voice command / app toggle
3. **Failure mode** (what happens if the internet is down?)
Example: light still works manually; automation may pause
4. **Device type** (bulb vs switch vs relay)
5. **Power source** (inverter-backed or not)
6. **Safety note** (high load? wet area? call electrician?)

Do this for just 3 rooms first: living room, bedroom, kitchen/passage. When those work smoothly for a week, expand.

In Chapter 3, we’ll tackle the invisible dealbreaker: Wi-Fi design, mesh placement, and the “IoT lane” that keeps your smart home responsive instead of random. **d**